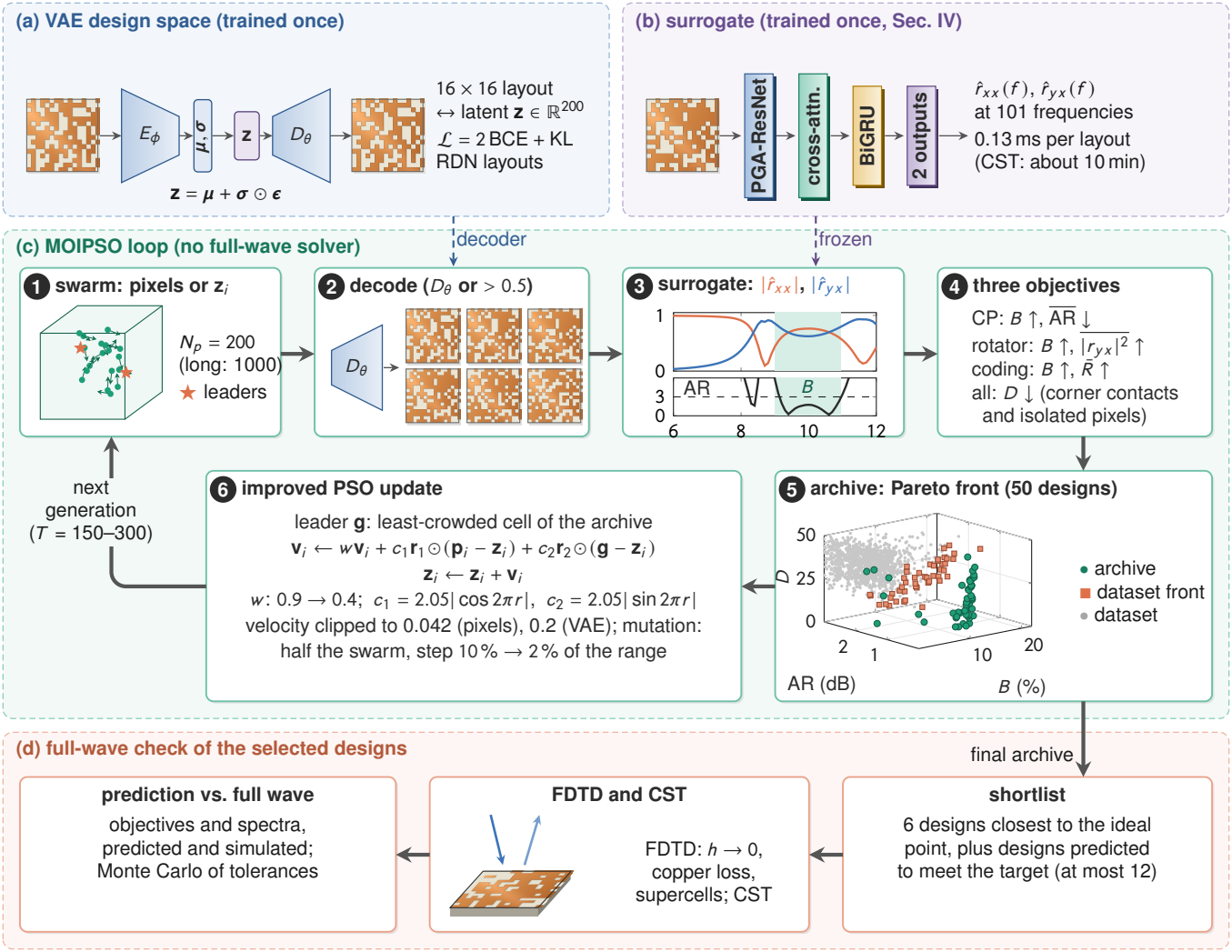

(a) VAE design space (trained once)
E_ϕ
μ, σ
z
D_θ
16 × 16 layout
↔ latent $\mathbf{z} \in \mathbb{R}^{200}$
$\mathcal{L}$ = 2 BCE + KL
RDN layouts
$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$
(b) surrogate (trained once, Sec. IV)
PGA-ResNet
cross-attn.
BiGRU
2 outputs
$\hat{r}_{xx}(f)$, $\hat{r}_{yx}(f)$
at 101 frequencies
0.13 ms per layout
(CST: about 10 min)
decoder
frozen
(c) MOIPSO loop (no full-wave solver)
1 swarm: pixels or $\mathbf{z}_i$
N_p = 200
(long: 1000)
★ leaders
2 decode (D_θ or > 0.5)
D_θ
3 surrogate: $|\hat{r}_{xx}|$, $|\hat{r}_{yx}|$
1
0
3
0
AR
B
6
8
10
12
4 three objectives
CP: $B \uparrow$, $\overline{\text{AR}} \downarrow$
rotator: $B \uparrow$, $|r_{yx}|^2 \uparrow$
coding: $B \uparrow$, $\bar{R} \uparrow$
all: $D \downarrow$ (corner contacts
and isolated pixels)
5 archive: Pareto front (50 designs)
50
25
0
D
2
1
10
20
AR (dB)
B (%)
archive
dataset front
dataset
6 improved PSO update
leader g: least-crowded cell of the archive
$\mathbf{v}_i \leftarrow w\mathbf{v}_i + c_1\mathbf{r}_1 \odot (\mathbf{p}_i - \mathbf{z}_i) + c_2\mathbf{r}_2 \odot (\mathbf{g} - \mathbf{z}_i)$
$\mathbf{z}_i \leftarrow \mathbf{z}_i + \mathbf{v}_i$
w: 0.9 → 0.4; $c_1 = 2.05|\cos 2\pi r|$, $c_2 = 2.05|\sin 2\pi r|$
velocity clipped to 0.042 (pixels), 0.2 (VAE); mutation:
half the swarm, step 10 % → 2 % of the range
next
generation
(T = 150–300)
final archive
(d) full-wave check of the selected designs
shortlist
6 designs closest to the ideal
point, plus designs predicted
to meet the target (at most 12)
FDTD and CST
FDTD: $h \to 0$,
copper loss,
supercells; CST
prediction vs. full wave
objectives and spectra,
predicted and simulated;
Monte Carlo of tolerances